

Changyi Li

Doctoral Researcher · School of Electrical Engineering (ELEC), Aalto University · advised by Yu Xiao

Doctoral researcher in computer vision, working on visual anomaly detection and cross-domain action recognition.

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RESEARCH INTERESTS

Visual anomaly detection (industrial & video) · cross-domain few-shot action recognition · multimodal sensing · large language models for structured data

EDUCATION & EXPERIENCE

- 2023 – now **Ph.D. Candidate, Computer Vision**
Aalto University, School of Electrical Engineering (ELEC) — Espoo, Finland
- 2022 – 2023 **Research Engineer**
Huawei, Chengdu Research Institute — Chengdu, China
- 2019 – 2022 **M.Sc.**
University of Electronic Science and Technology of China (UESTC) — Chengdu, China

HONORS & AWARDS

- 2026 **3rd place — Track 1 (Micro-Action Recognition); 2nd place — Track 2 (Multi-label Micro-Action Detection)**
3rd Micro-Action Analysis Grand Challenge (MAC 2026), in conjunction with ACM Multimedia 2026

PUBLICATIONS

PUBLISHED & ACCEPTED

- [1] **UniSLAD: A Unified Framework for Structural and Logical Industrial Visual Anomaly Detection**
Changyi Li, Chao Yang, Yu Xiao, Kari Tammi
IEEE CASE 2026
- [2] **Adaptive Multi-Granularity Temporal Modeling for Weakly Supervised Video Anomaly Detection**
Changyi Li, Yu Xiao
PRCV 2026
- [3] **Training-Free Logical and Structural Anomaly Detection via Calibrated Fusion**
Changyi Li, Miao Yu, Kai Dong, Yu Xiao
PRCV 2026
- [4] **Multi-Modal Sensing-Assisted Cross-Environment Beam Prediction Towards 6G**
Lei Wang, Changyi Li, Kai Dong, Sergiy A. Vorobyov, Esa Ollila
IEEE PIMRC 2026 · **Oral**
- [5] **A Portable Hybrid Quantum–Classical Neural Network for Continual Learning**
Zeyu Yang, Yunpeng Bai, Changyi Li, Yu Xiao
IEEE QCNC 2026

- [6] Exploring Human–AI Collaboration in E-Textile Design: A Case Study on Flex Sensor Placement for Shoulder Motion Detection
Zhuchenyang Liu, Yao Zhang, Yalan He, Hilla Paasio, **Changyi Li**, Guna Semjonova, Yu Xiao
Proc. ACM Hum.-Comput. Interact. 2026
- [7] Enhancing Knowledge Graph Interactions: A Comprehensive Text-to-Cypher Pipeline with Large Language Models
Chao Yang, **Changyi Li**, Xiaodu Hu, Hao Yu, Jinzhi Lu
Information Processing & Management 2025 · Journal
- [8] Revisiting the Impact of Domain Similarity on the Performance of Cross-Domain Few-Shot Activity Recognition
Changyi Li, Yu Xiao
IEEE BigData 2025
- [9] S2TKD: Dual-Student Knowledge Distillation for Industrial Visual Anomaly Detection and Localization
Changyi Li, Chao Yang, Yu Xiao
IEEE BigData 2025 · Oral
- PREPRINTS & UNDER REVIEW
- [10] COMPASS: Scene-Adaptive Contextual Modeling for Video Anomaly Detection
Changyi Li, Chao Yang, Yu Xiao, Kari Tammi
NeurIPS 2026 · under review
- [11] Diffusion Features Complement Vision–Language Models for Fine-Grained Visual Discrimination
Ziyu Hu, **Changyi Li**, Yu Xiao
NeurIPS 2026 · under review
- [12] CLAD: Complexity- and Logic-Aware Decomposition for Text-to-Cypher over Domain-Specific Knowledge Graphs
Chao Yang, **Changyi Li**, Ying Liu, Shen Yin, Peixing Zhang, Bing Zhu, Risto Ojala, Kari Tammi
EMNLP 2026 · under review
- [13] Reachability Is Not Generalization: Understanding Verb–Noun Decomposition in Assembly Action Recognition
Changyi Li, Yu Xiao
BMVC 2026 · under review
- [14] Closing the Estimation Gap: Training-Free Transductive Inference for Cross-Domain Few-Shot Action Recognition
Changyi Li, Yu Xiao
BMVC 2026 · under review
- [15] Compressed LLM Preprogramming for Vision-Aided Beam Prediction in Vehicular Networks
Kai Dong, Lei Wang, **Changyi Li**, Sergiy A. Vorobyov, Stefan Werner
IEEE Communications Letters · under review